Exercise 4: A Python-based address book

Let's use Python to construct a temporary address book, or contact list, which (for each entry) can contain a person's name, mobile number, house name / number, and postcode, all as strings.

1. Write Python code which creates an empty dictionary called "contacts". Create a function which takes a name, number, house name and postcode, and adds them to the "contacts" dictionary, using the name as the key (which is the best way to store the rest of the details, as the key's value?).

2. Write a while loop which will run until a user enters the word "quit", and within that loop fetch some input from the user. If user hasn't typed "quit", fetch a new name, number, house name and postcode from them, and use those details to add to the dictionary. Make you code print out all the names in "contacts" after each addition. You should now have a program which will keep asking the user for new contacts to enter in the address book, and which will quit when the user wishes.

3. Change your code so that the user may type "search" instead of "quit". If they do, have your program ask them to enter a name to search for. If the name they enter exists in the dictionary, print out that contact's details. If not, print a message to let the user know the person has not been found.

4. Try adapting your code so that the user must enter "add" if they wish to add an entry, and if they do not enter one of the words "quit", "search" or "add" the program will keep looping.

5. Optionally, you could add a feature which allows a user to search for a contact entry by mobile number. How can you adapt your code to achieve this easily?

Will this be quicker, slower, or the same as searching for a contact by name? What could you do (if anything) to change that?